Experiment No. 1 – Data Cleaning

Aim

To handle **missing values** in a dataset and identify and handle **outliers using the Z-score method**.

Python Program

```
import pandas as pd
import numpy as np
from scipy import stats

# Create a sample dataset
data = {
    'Name': ['A', 'B', 'C', 'D', 'E', 'F', 'G'],
    'Age': [21, 22, np.nan, 23, 24, 100, 22],
    'Salary': [25000, 28000, 30000, np.nan, 27000, 200000, 29000]
}

df = pd.DataFrame(data)

print("Original Dataset:")
print(df)

# ----------------------------------
# a) HANDLING MISSING VALUES
# ----------------------------------

# Replace missing values with the mean
df['Age'] = df['Age'].fillna(df['Age'].mean())
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())

print("\nDataset after handling missing values:")
print(df)

# ----------------------------------
# b) IDENTIFYING AND HANDLING OUTLIERS
```

```
# ----------------------------------

# Select numerical columns
numerical_columns = ['Age', 'Salary']

# Calculate Z-score
z_scores = np.abs(stats.zscore(df[numerical_columns]))

# Keep rows having Z-score less than 3
cleaned_df = df[(z_scores < 3).all(axis=1)]

print("\nDataset after removing outliers:")
print(cleaned_df)
```

Explanation

a) Handling Missing Values:
Missing values (`NaN`) in the **Age** and **Salary** columns are replaced using their respective **mean values**. This technique is called **imputation**.

b) Handling Outliers:
The **Z-score method** is used to identify outliers. A data value with a Z-score greater than **3** is generally considered an outlier. Such rows are removed from the dataset.

Conclusion

Thus, the dataset was successfully cleaned by handling missing values using **mean imputation** and detecting/removing outliers using the **Z-score method**.

```
Original Dataset:
  Name    Age    Salary
0    A   21.0   25000.0
1    B   22.0   28000.0
2    C    NaN   30000.0
3    D   23.0       NaN
4    E   24.0   27000.0
5    F  100.0  200000.0
6    G   22.0   29000.0

Dataset after handling missing values:
  Name         Age    Salary
0    A   21.000000   25000.0
1    B   22.000000   28000.0
2    C   35.333333   30000.0
3    D   23.000000   56500.0
4    E   24.000000   27000.0
5    F  100.000000  200000.0
6    G   22.000000   29000.0

Dataset after removing outliers:
  Name         Age    Salary
0    A   21.000000   25000.0
1    B   22.000000   28000.0
2    C   35.333333   30000.0
3    D   23.000000   56500.0
4    E   24.000000   27000.0
5    F  100.000000  200000.0
6    G   22.000000   29000.0
```